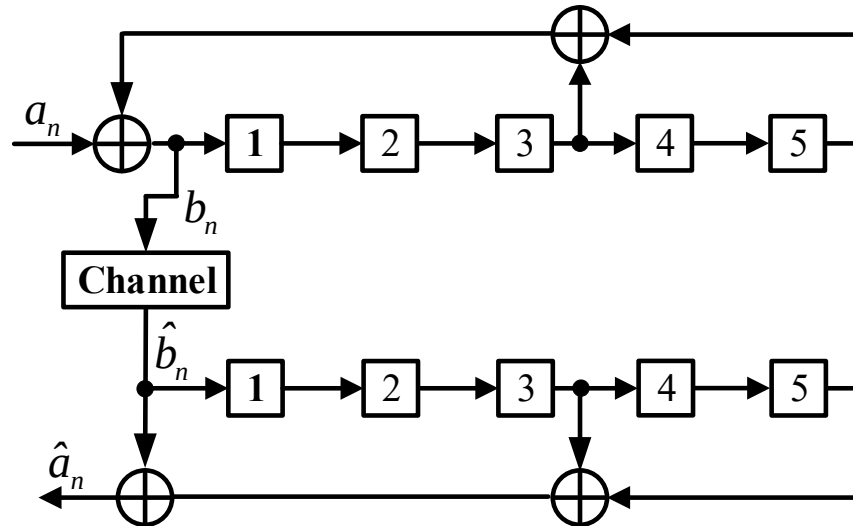


Problem:

Design a Self-Synchronization Scrambler/Descrambler

Design Description:

1. The pseudorandom polynomial $f(x) = x^5 + x^3 + 1$
2. The block diagram of the self-synchronization scrambler/descrambler is shown as below.



3. The initial values of $\{b_{n-1}, b_{n-2}, b_{n-3}, b_{n-4}, b_{n-5}\}$ and $\{\hat{b}_{n-1}, \hat{b}_{n-2}, \hat{b}_{n-3}, \hat{b}_{n-4}, \hat{b}_{n-5}\}$ are equivalent to $\{1, 1, 1, 1, 1\}$ and $\{0, 0, 1, 0, 0\}$, respectively.
4. The format of the input sequences is described as $a_n = [5, 0, A, 5, 1, A, 5, 2, A, 5, 3, A, \dots]$, where all values in a_n are hexadecimal.
5. The channel is assumed to be ideal, namely, $h_n = 1$.
6. Finally, you have to submit your m-file to E-Learning and your hard-copy report at class.